STPM Physics · Semester 1 · Chapter 8

STATICS

8.1 Equilibrium of Forces

DEFINITIONS & LAWS

- **Translational equilibrium**: a body is in translational equilibrium when the resultant force acting on it is zero; it is at rest or moves with constant velocity.
- **Rotational equilibrium**: a body is in rotational equilibrium when the resultant torque about any axis is zero.
- **Conditions for equilibrium of a rigid body (static equilibrium)**: (1) the vector sum of all forces is zero, and (2) the sum of the moments about any point is zero.
- **Triangle of forces**: if three coplanar forces acting at a point are in equilibrium, they can be represented in magnitude and direction by the three sides of a triangle taken in order.
- For three non-parallel forces in equilibrium, the lines of action are coplanar and meet at one point (concurrent).
- A couple is two equal and opposite parallel forces; a body acted on by a couple alone is in translational equilibrium but not in rotational equilibrium.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Translational equilibrium	$\Sigma F = 0$; $\Sigma F_x = 0$ and $\Sigma F_y = 0$	ΣF = resultant force (N); F_x, F_y = components along x and y axes; for coplanar forces
Rotational equilibrium	$\Sigma \tau = 0$	τ = torque (N m); about any chosen point; Σ clockwise moments = Σ anticlockwise moments
Lami's theorem	$\frac{F_1}{\sin \alpha} = \frac{F_2}{\sin \beta} = \frac{F_3}{\sin \gamma}$	F_1, F_2, F_3 = three forces in equilibrium (N); α, β, γ = angles opposite F_1, F_2, F_3 (the angles between the other two forces)
Resolving a force	$F_x = F\cos\theta$; $F_y = F\sin\theta$	F = force (N); θ = angle of F to the horizontal; F_x = horizontal component (N); F_y = vertical component (N)
Weight	$W = mg$	W = weight (N); m = mass (kg); g = gravitational field strength (N kg^{-1}); acts vertically downwards through the centre of gravity
Friction at a rough surface	$f \leq \mu N$; $f = \mu N$ (limiting)	f = friction force (N); μ = coefficient of friction; N = normal reaction (N); equality holds at the limiting value

TIPS TO ANSWER

- Draw a clear free-body diagram showing every force (weight, tension, normal reaction, friction) with arrows and angles.
- Resolve horizontally and vertically, write $\Sigma F_x = 0$ and $\Sigma F_y = 0$, solve; state the answer with unit and direction.
- For a rigid body, take moments about the point where the most unknown forces act so that they have zero moment, and state "Σ clockwise moments = Σ anticlockwise moments".
- Use the triangle of forces/Lami's theorem for exactly three forces; the closed triangle must be drawn head-to-tail.

SECTION A TRAPS

Trap: Zero resultant force does not guarantee zero resultant torque (a couple); both conditions are needed for static equilibrium.

Trap: Using the wrong angle in resolution: when the angle is given with the vertical, the horizontal component is F sinθ, not F cosθ.

Trap: In Lami's theorem the sine is of the angle opposite that force (the angle between the other two forces), not the angle the force makes with the vertical.

Trap: Tension is not equal to the weight when the string is at an angle; T cosθ = W for two symmetric strings gives $T = \frac{W}{2\cos\theta}$.

TRICKS

Trick: Check: if two forces are equal and opposite and colinear, the third force must be zero; for three forces the largest force is opposite the largest angle.

Trick: Taking moments about the pivot removes the pivot's unknown reaction from the equation; saves one simultaneous equation.

8.2 Moments and Centre of Gravity

DEFINITIONS & LAWS

- **Moment of a force** about a point is the product of the force and the perpendicular distance from the point to the line of action of the force.

- **Principle of moments**: for a body in rotational equilibrium, the sum of the clockwise moments about any point is equal to the sum of the anticlockwise moments about the same point.
- **Centre of gravity** of a body is the point through which the whole weight of the body appears to act.
- **Centre of mass** is the point at which the whole mass of the body can be considered to be concentrated; it coincides with the centre of gravity when the gravitational field is uniform.
- The centre of gravity of a uniform regular body is at its geometric centre; for a freely suspended body it lies vertically below the point of suspension.
- **Stable equilibrium**: a body returns to its original position after a small displacement; the centre of gravity rises on displacement. **Unstable**: the body moves further away; the centre of gravity falls. **Neutral**: the body stays in the new position; the centre of gravity stays at the same height.
- A body topples when the vertical line through its centre of gravity falls outside its base.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Moment of a force	Moment = Fd	F = force (N); d = perpendicular distance to the line of action (m); moment in N m
Principle of moments	ΣF d (clockwise) = ΣF d (anticlockwise)	F = force (N); d = perpendicular distance (m); about the same point
Centre of gravity of a system of masses	$\bar{x} = \frac{\Sigma mx}{\Sigma m}$	$\bar{x}$ = position of the centre of gravity (m); m = mass (kg); x = position of mass m (m)
Uniform beam	W = mg	W = weight (N); m = mass (kg); g = gravitational field strength (N kg^{-1}); weight acts at the midpoint
Toppling on a slope	$\tan\theta > \frac{w}{h}$; $\tan\theta = \frac{\frac{w}{2}}{\frac{h}{2}}$	Uniform block of base width w (m) and height h (m) on a slope of angle θ; topples when $\tan\theta > \frac{w}{h}$; the line through the centre of gravity passes the lower edge at $\tan\theta = \frac{\frac{w}{2}}{\frac{h}{2}}$
Toppling on a horizontal surface	$P\,h \geq W\,\frac{w}{2}$	P = horizontal force (N) applied at height h (m); W = weight (N); w = base width (m); block topples when this holds
Reaction forces on a beam	$R_1 + R_2 = W$ + other loads	R_1, R_2 = reactions at the two supports (N); W = weight of beam (N); translational equilibrium

TIPS TO ANSWER

- State the pivot chosen, then "taking moments about P" and write clockwise = anticlockwise with each force × perpendicular distance.

- Mark the weight of a uniform beam at its midpoint, and the reaction at each support; show the distances on the diagram.
- Combine $\Sigma F = 0$ with moments to find unknown reactions or tensions; give each answer in N with the sense stated.
- For stability, write that the vertical line through the centre of gravity falls inside/outside the base; mention a low centre of gravity and a wide base increase stability.

SECTION A TRAPS

Trap: Using the length along the beam as the distance for an inclined force; the perpendicular distance (or $F \sin\theta$) must be used.

Trap: Forgetting the weight of the uniform beam itself at its midpoint when taking moments.

Trap: Stability: a lower centre of gravity and a wider base increase stability; the mass of the object does not by itself decide whether it topples.

Trap: Pivot at the wrong point in the equation: all moments in one equation must be about the same point; the force at the pivot has zero moment.

TRICKS

Trick: Choose the pivot where an unknown force acts to eliminate that force from the equation.

Trick: Block on a slope: the critical angle satisfies $\tan\theta = \frac{w}{h}$, so a tall narrow block (small $\frac{w}{h}$) topples at a smaller angle.